

Project #1: Interactive Calculator

Performing Basic Mathematical Operations

Description:

In this project, students will write a Python program that:

- Prompts the user to choose a mathematical operation.
- Takes two numbers as input.
- Performs one of the following operations:
 - Addition
 - Subtraction
 - Multiplication
 - Division
 - Modulus
 - Exponentiation
- Handles division or modulus by zero using simple if conditions (no loops, no error handling).
- Displays the final result clearly.

This project helps students understand how decisions are made in programs using if-elif-else logic.

Algorithm:

Step 1: Show Menu

- Display the list of available operations for the user:
 - Add
 - Subtract
 - Multiply
 - Divide
 - Modulus
 - Exponentiation
- Ask the user to enter their choice.

Step 2: Input

- Ask the user to enter two numbers.
- Store them in variables `num1` and `num2`.
- Convert the inputs to float.

Step 3: Perform the Operation

Based on the user's choice:

- **Addition:** $\text{result} = \text{num1} + \text{num2}$
- **Subtraction:** $\text{result} = \text{num1} - \text{num2}$
- **Multiplication:** $\text{result} = \text{num1} \times \text{num2}$
- **Division:**
 - If $\text{num2} == 0$, display "Cannot divide by zero!"
 - Else perform division
- **Modulus:**
 - If $\text{num2} == 0$, display "Cannot divide by zero!"
 - Else $\text{result} = \text{num1} \% \text{num2}$
- **Exponentiation:** $\text{result} = \text{num1} ** \text{num2}$
- If the choice does not match any option, print "Invalid choice".

Step 4: Output

- Display the result of the operation.
- Print a final thank-you message.

Pseudocode:

```
BEGIN

PRINT "Welcome to Your Awesome Calculator!"
PRINT "1. Add"
PRINT "2. Subtract"
PRINT "3. Multiply"
PRINT "4. Divide"
PRINT "5. Modulus"
PRINT "6. Exponentiation"

INPUT choice

INPUT num1
INPUT num2

IF choice == 1 THEN
    result = num1 + num2
    PRINT result

ELSE IF choice == 2 THEN
    result = num1 - num2
    PRINT result

ELSE IF choice == 3 THEN
    result = num1 * num2
    PRINT result

ELSE IF choice == 4 THEN
    IF num2 == 0 THEN
        PRINT "Cannot divide by zero!"
    ELSE
        result = num1 / num2
        PRINT result
    END IF

ELSE IF choice == 5 THEN
    IF num2 == 0 THEN
```

```

        PRINT "Cannot divide by zero!"
    ELSE
        result = num1 % num2
        PRINT result
    END IF

ELSE IF choice == 6 THEN
    result = num1 ** num2
    PRINT result

ELSE
    PRINT "Invalid choice"
END IF

PRINT "Thanks for using the Awesome Calculator!"

END

```

Sample Output:

Example 1

Welcome to Your Awesome Calculator!

1. Add
2. Subtract
3. Multiply
4. Divide
5. Modulus
6. Exponentiation

Enter your choice: 1

Enter first number: 5

Enter second number: 3

Your answer is: 8

Thanks for using the Awesome Calculator!

Example 2

Enter your choice: 4

Enter first number: 9

Enter second number: 0

Cannot divide by zero!

Thanks for using the Awesome Calculator!

Example 3

Enter your choice: 6

Enter first number: 2

Enter second number: 5

Your answer is: 32

Thanks for using the Awesome Calculator!

Learning Outcome:

- How to take user input in Python.
- How to convert values to numeric types (float).
- How to use if-elif-else to control program flow.
- How to implement basic math operations using Python operators.
- How to manually handle division or modulus by zero without using exceptions.
- How menu-driven programs work.